

Online Appendix: Large Hybrid Time-Varying Parameter VARs

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Appendix A: Estimation Details

In this appendix we provide estimation details of the hybrid TVP-VAR. In particular, we describe the details of Step 2 - Step 7 of the posterior sampler outlined in Section 3.2 of the main text.

Step 2. To sample $\mathbf{h}_i$, the log-volatility vector of the i -th equation, $i = 1, \dots, n$, we can use the auxiliary mixture sampler of Kim, Shephard, and Chib (1998). More specifically, we first compute the residuals $\varepsilon_{i,t}^y, t = 1, \dots, T$. Then, we transform these residuals as $\mathbf{y}_i^* = (\log(\varepsilon_{i,1}^y)^2, \dots, \log(\varepsilon_{i,T}^y)^2)'$. Finally, we implement the auxiliary mixture sampler in conjunction with the precision sampler of Chan and Jeliazkov (2009) to sample $\mathbf{h}_i$ using $\mathbf{y}_i^*$ as data.

Step 3. The parameters $\Sigma_{\theta_i}^{\frac{1}{2}} = \text{diag}(\Sigma_{\beta_i}^{\frac{1}{2}}, \Sigma_{\alpha_i}^{\frac{1}{2}})$ and $\theta_{i,0}$ can be sampled easily as their joint distribution is Gaussian. To see that, let $\boldsymbol{\mu}_i^\theta = (\theta'_{i,0}, \sigma_{\beta_i,1}, \dots, \sigma_{\beta_i,k_\beta}, \sigma_{\alpha_i,1}, \dots, \sigma_{\alpha_i,k_{\alpha_i}})'$ and define $\mathbf{w}_{i,t}^\theta = (\mathbf{x}_{i,t}, \gamma_i^\beta \tilde{\mathbf{x}}_t \odot \tilde{\boldsymbol{\beta}}_{i,t}, \gamma_i^\alpha \tilde{\mathbf{w}}_{i,t} \odot \tilde{\boldsymbol{\alpha}}_{i,t})$, where $\odot$ denotes the component-wise product.

Then, we can rewrite the i -th equation as a linear regression:

$$y_{i,t} = \mathbf{w}_{i,t}^\theta \boldsymbol{\mu}_i^\theta + \varepsilon_{i,t}^y.$$

Since both $\Sigma_{\theta_i}^{\frac{1}{2}}$ and $\boldsymbol{\theta}_{i,0}$ have Gaussian priors, the implied prior on $\boldsymbol{\mu}_i^\theta$ is also Gaussian: $\boldsymbol{\mu}_i^\theta \sim \mathcal{N}(\mathbf{0}, \mathbf{V}_{\boldsymbol{\mu}_i^\theta})$, where $\mathbf{V}_{\boldsymbol{\mu}_i^\theta} = \text{diag}(\mathbf{V}_{\boldsymbol{\theta}_{i,0}}, S_{\theta_{i,1}}, \dots, S_{\theta_{i,k_{\theta_i}}})$. Define $\mathbf{W}_i^\theta$ by stacking $\mathbf{w}_{i,t}^\theta$ over $t = 1, \dots, T$. It follows that the full conditional distribution of $\boldsymbol{\mu}_i^\theta$ is given by

$$(\boldsymbol{\mu}_i^\theta \mid \mathbf{y}_i, \tilde{\boldsymbol{\theta}}_i, \mathbf{h}_i, \boldsymbol{\gamma}_i) \sim \mathcal{N}(\hat{\boldsymbol{\mu}}_i^\theta, \mathbf{K}_{\boldsymbol{\mu}_i^\theta}^{-1}),$$

where $\mathbf{K}_{\boldsymbol{\mu}_i^\theta} = \mathbf{V}_{\boldsymbol{\mu}_i^\theta}^{-1} + (\mathbf{W}_i^\theta)' \boldsymbol{\Omega}_{\mathbf{h}_i}^{-1} \mathbf{W}_i^\theta$ and $\hat{\boldsymbol{\mu}}_i^\theta = \mathbf{K}_{\boldsymbol{\mu}_i^\theta}^{-1} (\mathbf{W}_i^\theta)' \boldsymbol{\Omega}_{\mathbf{h}_i}^{-1} \mathbf{y}_i$ with $\boldsymbol{\Omega}_{\mathbf{h}_i} = \text{diag}(e^{h_{i,1}}, \dots, e^{h_{i,T}})$.

Steps 4-5. The full conditional distributions of $\sigma_{h,i}^2$ and $h_{i,0}$ are standard and they can be sampled easily. In particular, their full conditional distributions are

$$\begin{aligned} (\sigma_{h,i}^2 \mid \mathbf{h}_i, h_{i,0}) &\sim \mathcal{IG} \left(\nu_{h,i} + \frac{T}{2}, S_{h,i} + \frac{1}{2} \sum_{t=1}^T (h_{i,t} - h_{i,t-1})^2 \right), \\ (h_{i,0} \mid \mathbf{h}_i, \sigma_{h,i}^2) &\sim \mathcal{N}(\hat{h}_{i,0}, K_{h_{i,0}}^{-1}), \end{aligned}$$

where $K_{h_{i,0}} = 1/V_{h_{i,0}} + 1/\sigma_{h,i}^2$ and $\hat{h}_{i,0} = K_{h_{i,0}}^{-1} (a_{h_{i,0}}/V_{h_{i,0}} + h_{i,1}/\sigma_{h,i}^2)$.

Step 6. Next, given the independent beta priors on p_i^β and $p_i^\alpha, i = 1, \dots, n$ their full conditional posterior distributions are also beta distributions. In fact, we have:

$$\begin{aligned} (p_i^\beta \mid \gamma_i^\beta) &\sim \mathcal{B}(a_{p^\beta} + \gamma_i^\beta, b_{p^\beta} + 1 - \gamma_i^\beta), \\ (p_i^\alpha \mid \gamma_i^\alpha) &\sim \mathcal{B}(a_{p^\alpha} + \gamma_i^\alpha, b_{p^\alpha} + 1 - \gamma_i^\alpha). \end{aligned}$$

Step 7. To implement Step 7, we follow the sampling approach in Chan (2021). First

note that κ_1 and κ_2 only appear in their priors $\kappa_j \sim \mathcal{G}(c_{1,j}, c_{2,j})$, $j = 1, 2$, and in the prior covariance matrices $\mathbf{V}_{\boldsymbol{\theta}_{i,0}}$, $i = 1, \dots, n$. Letting $\theta_{ij,0}$ denote the j -th element of $\boldsymbol{\theta}_{i,0}$, we define the index set S_{κ_1} to be the collection of indexes (i, j) such that $\theta_{ij,0}$ is a coefficient associated with an own lag. That is, $S_{\kappa_1} = \{(i, j) : \theta_{ij,0} \text{ is a coefficient associated with an own lag}\}$. Similarly, define S_{κ_2} as the set that collects all the indexes (i, j) such that $\theta_{ij,0}$ is a coefficient associated with a lag of other variables. It is easy to check that the numbers of elements in S_{κ_1} and S_{κ_2} are respectively np and $(n-1)np$. Furthermore, for $(i, j) \in S_{\kappa_1} \cup S_{\kappa_2}$, let

$$C_{ij} = \begin{cases} \frac{1}{l^2}, & \text{for the coefficient on the } l\text{-th lag of variable } i, \\ \frac{s_i^2}{l^2 s_j^2}, & \text{for the coefficient on the } l\text{-th lag of variable } j, j \neq i. \end{cases}$$

Then, we have

$$\begin{aligned} p(\kappa_1 | \boldsymbol{\theta}_0) &\propto \prod_{(i,j) \in S_{\kappa_1}} \kappa_1^{-\frac{1}{2}} e^{-\frac{1}{2\kappa_1 C_{ij}} \theta_{ij,0}^2} \times \kappa_1^{c_{1,1}-1} e^{-\kappa_1 c_{2,1}} \\ &= \kappa_1^{c_{1,1} - \frac{np}{2} - 1} e^{-\frac{1}{2} \left(2c_{2,1} \kappa_1 + \kappa_1^{-1} \sum_{(i,j) \in S_{\kappa_1}} \frac{\theta_{ij,0}^2}{C_{ij}} \right)}, \end{aligned}$$

which is the kernel of the generalized inverse Gaussian distribution:

$$(\kappa_1 | \boldsymbol{\theta}_0) \sim \mathcal{GIG} \left(c_{1,1} - \frac{np}{2}, 2c_{2,1}, \sum_{(i,j) \in S_{\kappa_1}} \frac{\theta_{ij,0}^2}{C_{ij}} \right).$$

Similarly, κ_2 also has a generalized inverse Gaussian distribution:

$$(\kappa_2 | \boldsymbol{\theta}_0) \sim \mathcal{GIG} \left(c_{1,2} - \frac{(n-1)np}{2}, 2c_{2,2}, \sum_{(i,j) \in S_{\kappa_2}} \frac{\theta_{ij,0}^2}{C_{ij}} \right).$$

Appendix B: Technical Details on Model Comparison

This appendix outlines the technical details on comparing hybrid TVP-VARs using the Savage-Dickey density ratio. First, suppose we want to compare the proposed hybrid TVP-VAR with γ unrestricted against a TVP-VAR characterized by $\gamma = \mathbf{c} \in \{0, 1\}^{2n}$. Then, the Bayes factor in favor of the unrestricted model can be expressed as the Savage-Dickey density ratio

$$\text{BF}_{\mathbf{u}, \mathbf{c}} = \frac{p(\gamma = \mathbf{c})}{p(\gamma = \mathbf{c} | \mathbf{y})},$$

provided that the priors under the restricted and unrestricted models satisfy a compatibility condition (see, e.g., Verdinelli and Wasserman, 1995, for details). The priors described in Section 3.1 of the main text assume that the restricted (i.e., γ) and unrestricted parameters are independent *a priori*, which implies the compatibility condition.

Next, we outline how one can evaluate the marginal prior and posterior densities at $\gamma = \mathbf{c}$. First, to evaluate $p(\gamma = \mathbf{c})$, recall that under our priors, γ_i^β and γ_i^α follow independent Bernoulli distributions with success probabilities p_i^β and p_i^α , respectively, for $i = 1, \dots, n$. These success probabilities, in turn, are assumed to have beta distributions: $p_i^\beta \sim \mathcal{B}(a_{p^\beta}, b_{p^\beta})$ and $p_i^\alpha \sim \mathcal{B}(a_{p^\alpha}, b_{p^\alpha})$. Hence, the marginal prior of γ_i^β (unconditional on p_i^β) can be computed as

$$\begin{aligned} p(\gamma_i^\beta) &= \int_0^1 (p_i^\beta)^{\gamma_i^\beta} (1 - p_i^\beta)^{1-\gamma_i^\beta} \times \frac{\Gamma(a_{p^\beta} + b_{p^\beta})}{\Gamma(a_{p^\beta})\Gamma(b_{p^\beta})} (p_i^\beta)^{a_{p^\beta}-1} (1 - p_i^\beta)^{b_{p^\beta}-1} dp_i^\beta \\ &= \frac{\Gamma(a_{p^\beta} + b_{p^\beta})\Gamma(\gamma_i^\beta + a_{p^\beta})\Gamma(1 - \gamma_i^\beta + b_{p^\beta})}{\Gamma(a_{p^\beta})\Gamma(b_{p^\beta})\Gamma(a_{p^\beta} + b_{p^\beta} + 1)}, \end{aligned}$$

where $\Gamma(\cdot)$ is the gamma function. A similar expression can be derived for $p(\gamma_i^\alpha)$. It

follows that the marginal prior for $\boldsymbol{\gamma}$ has the following analytical expression

$$\begin{aligned} p(\boldsymbol{\gamma}) &= \prod_{i=1}^n p(\gamma_i^\beta) p(\gamma_i^\alpha) \\ &= \prod_{i=1}^n \frac{\Gamma(a_{p^\beta} + b_{p^\beta}) \Gamma(\gamma_i^\beta + a_{p^\beta}) \Gamma(1 - \gamma_i^\beta + b_{p^\beta})}{\Gamma(a_{p^\beta}) \Gamma(b_{p^\beta}) \Gamma(a_{p^\beta} + b_{p^\beta} + 1)} \frac{\Gamma(a_{p^\alpha} + b_{p^\alpha}) \Gamma(\gamma_i^\alpha + a_{p^\alpha}) \Gamma(1 - \gamma_i^\alpha + b_{p^\alpha})}{\Gamma(a_{p^\alpha}) \Gamma(b_{p^\alpha}) \Gamma(a_{p^\alpha} + b_{p^\alpha} + 1)}, \end{aligned}$$

which can be easily evaluated at any point $\mathbf{c}$.

Next, we can evaluate $p(\boldsymbol{\gamma} = \mathbf{c} | \mathbf{y})$ with $\mathbf{c} = (c_{1,1}, c_{1,2}, \dots, c_{n,1}, c_{n,2})'$ using the Monte Carlo average:

$$\frac{1}{R} \sum_{r=1}^R \prod_{i=1}^n \mathbb{P} \left(\gamma_i = (c_{i,1}, c_{i,2}) | \mathbf{y}_i, \mathbf{h}_i^{(r)}, \boldsymbol{\Sigma}_{\theta_i}^{(r)}, \boldsymbol{\theta}_{i,0}^{(r)} \right),$$

where $\mathbf{h}_i^{(r)}, \boldsymbol{\Sigma}_{\theta_i}^{(r)}, \boldsymbol{\theta}_{i,0}^{(r)}, i = 1, \dots, n, r = 1, \dots, R$ are posterior draws from the unrestricted model. Analytical expressions of the above posterior probabilities are given in Section 3.2 of the main text. For numerical stability, both the prior and posterior densities are computed in log scale.

Thus, we have shown how one can compare the proposed hybrid TVP-VAR with $\boldsymbol{\gamma}$ unrestricted against any TVP-VAR characterized by $\boldsymbol{\gamma} = \mathbf{c} \in \{0, 1\}^{2n}$. To compare two TVP-VARs characterized by $\boldsymbol{\gamma} = \mathbf{c}_1$ and $\boldsymbol{\gamma} = \mathbf{c}_2$, note that one can express the Bayes factor in favor of $\boldsymbol{\gamma} = \mathbf{c}_1$ as

$$\text{BF}_{\mathbf{c}_1, \mathbf{c}_2} = \frac{\text{BF}_{\mathbf{u}, \mathbf{c}_2}}{\text{BF}_{\mathbf{u}, \mathbf{c}_1}} = \frac{p(\boldsymbol{\gamma} = \mathbf{c}_1 | \mathbf{y})}{p(\boldsymbol{\gamma} = \mathbf{c}_2 | \mathbf{y})}.$$

This expression can be evaluated using posterior draws from the unrestricted model as before.

Appendix C: Data

The dataset covers 20 quarterly variables sourced from the FRED-QD database at the Federal Reserve Bank of St. Louis (McCracken and Ng, 2021). The sample period is from 1959Q1 to 2018Q4. Table 1 lists all the variables and describes how they are transformed. For example, $\Delta \log$ is used to denote the first difference in the logs, i.e., $\Delta \log x = \log x_t - \log x_{t-1}$.

Table 1: Description of variables used in empirical application.

Variable	Mnemonic	Transformation
Real Gross Domestic Product	GDPC1	400 $\Delta \log$
Personal Consumption Expenditures: Chain-type Price index	PCECTPI	400 $\Delta \log$
Civilian Unemployment Rate	UNRATE	no transformation
Effective Federal Funds Rate	FEDFUNDS	no transformation
Industrial Production Index	INDPRO	400 $\Delta \log$
Real Average Hourly Earnings of Production and Nonsupervisory Employees: Manufacturing	CES30000000008x	400 $\Delta \log$
Real M1 Money Stock	M1REAL	400 $\Delta \log$
Real Personal Consumption Expenditures	PCECC96	400 $\Delta \log$
Real Disposable Personal Income	DPIC96	400 $\Delta \log$
Industrial Production: Final Products	IPFINAL	400 $\Delta \log$
All Employees: Total nonfarm	PAYEMS	400 $\Delta \log$
Civilian Employment	CE16OV	400 $\Delta \log$
Nonfarm Business Section: Hours of All Persons	HOANBS	400 $\Delta \log$
Gross Domestic Product: Chain-type Price index	GDPCTPI	400 $\Delta \log$
Consumer Price Index for All Urban Consumers: All Items	CPIAUCSL	400 $\Delta \log$
Producer Price Index for All commodities	PPIACO	400 $\Delta \log$
Nonfarm Business Sector: Real Compensation Per Hour	COMPRNFB	400 $\Delta \log$
Nonfarm Business Section: Real Output Per Hour of All Persons	OPHNFB	400 $\Delta \log$
10-Year Treasury Constant Maturity Rate	GS10	no transformation
Real M2 Money Stock	M2REAL	400 $\Delta \log$

Appendix D: Additional Results

In this appendix we provide additional simulation and empirical results. First, we investigate the effect of the prior on γ_i^β and γ_i^α by replacing the beta prior $\mathcal{B}(0.5, 0.5)$ by the uniform prior $\mathcal{U}(0, 1)$ in the Monte Carlo experiments. Table 2 reports the frequencies of the posterior modes of γ_i^β and γ_i^α being one in 300 datasets. All in all, the simulation results are similar to the baseline case, suggesting that the posterior estimates are not sensitive to the prior on γ_i^β and γ_i^α .

Table 2: Frequencies (%) of the posterior modes of γ_i^β and γ_i^α being one in 300 datasets with $T = 400$. The two sets of priors for γ_i^β and γ_i^α are $\mathcal{B}(0.5, 0.5)$ and $\mathcal{U}(0, 1)$ priors.

Equation	True γ_i^β	True γ_i^α	$\mathcal{B}(0.5, 0.5)$ prior		$\mathcal{U}(0, 1)$ prior	
			γ_i^β	γ_i^α	γ_i^β	γ_i^α
1	0	0	0.06	—	0.06	—
2	0	1	0.04	0.88	0.03	0.87
3	1	0	0.98	0.25	0.99	0.24
4	1	1	0.98	0.64	0.98	0.62
5	0	0	0.02	0.02	0.00	0.02
6	0	1	0.03	0.96	0.02	0.96
7	1	0	0.97	0.13	0.98	0.11
8	1	1	0.95	0.80	0.95	0.81
9	0	0	0.03	0.00	0.01	0.00
10	0	1	0.04	0.94	0.02	0.96
11	1	0	0.94	0.11	0.97	0.10
12	1	1	0.93	0.88	0.93	0.90

Next, we compute the inefficiency factors of the posterior draws from the 20-variable hybrid TVP-VAR in Section 5 of the main text, defined as

$$1 + 2 \sum_{l=1}^L \rho_l,$$

where ρ_l is the sample autocorrelation at lag length l and L is chosen to be large enough

so that the autocorrelation tapers off. In the ideal case where the posterior draws are independent, the corresponding inefficiency factor is 1. Figure 1 reports the inefficiency factors, obtained using 10,000 posterior draws after a burn-in period of 1,000. The results show that the proposed posterior sampler is efficient in terms of producing posterior draws that are not highly autocorrelated.

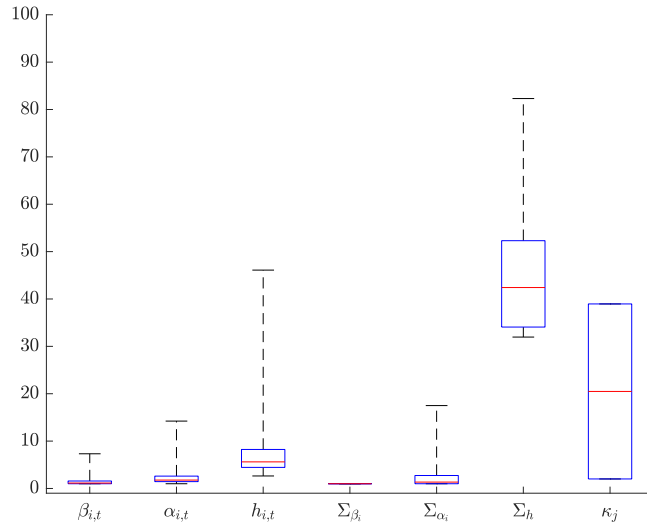


Figure 1: Boxplots of inefficiency factors of posterior draws from the 20-variable hybrid TVP-VAR.

Finally, we report additional forecasting results. First, Figure 2 reports forecasting results of the full-fledged TVP-VAR HYB-(1,1) relative to the benchmark HYB-(1,0), a TVP-VAR with stochastic volatility and a constant impact matrix (by setting all γ_i^β to 1 and all γ_i^α to 0). The results show that the forecast performance of the two models are mostly similar.

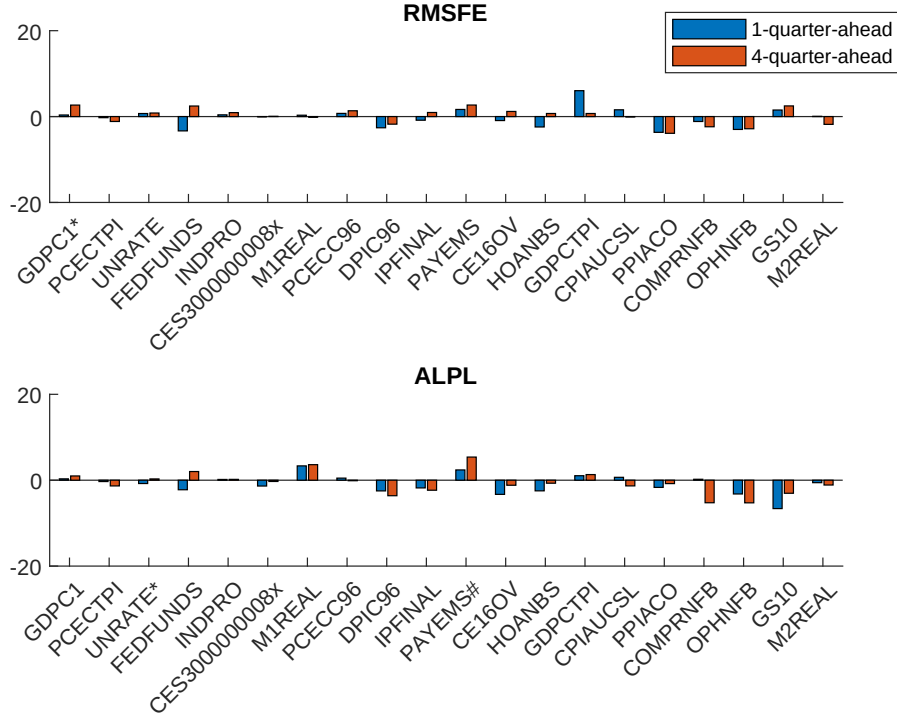


Figure 2: Forecasting results of HYB-(1, 1) compared to the benchmark HYB-(1, 0). The top panel shows the percentage gains in root mean squared forecast error of HYB-(1, 1). The symbols * and # after the mnemonic indicate rejection of equal forecast accuracy at significance level 0.05 using the test in Diebold and Mariano (1995) for 1- and 4-quarter-ahead point forecasts, respectively. The bottom panel presents the percentage gains in the average of log predictive likelihoods.

Next, Figure 3 reports the forecasting results of an extension of the hybrid TVP-VAR where the innovations to $h_{i,t}$ are correlated across equations relative to the benchmark HYB where the innovations are independent. The results show that the forecast performance of the extension is on average better in both point and density forecasts, although the forecast gains are modest. For example, the median percentage gains in RMSFE for 1- and 4-quarter-ahead forecasts are 0.6% and 0.4%, respectively; the median percentage gains in ALPL are 1.5% and 0.6%, respectively.

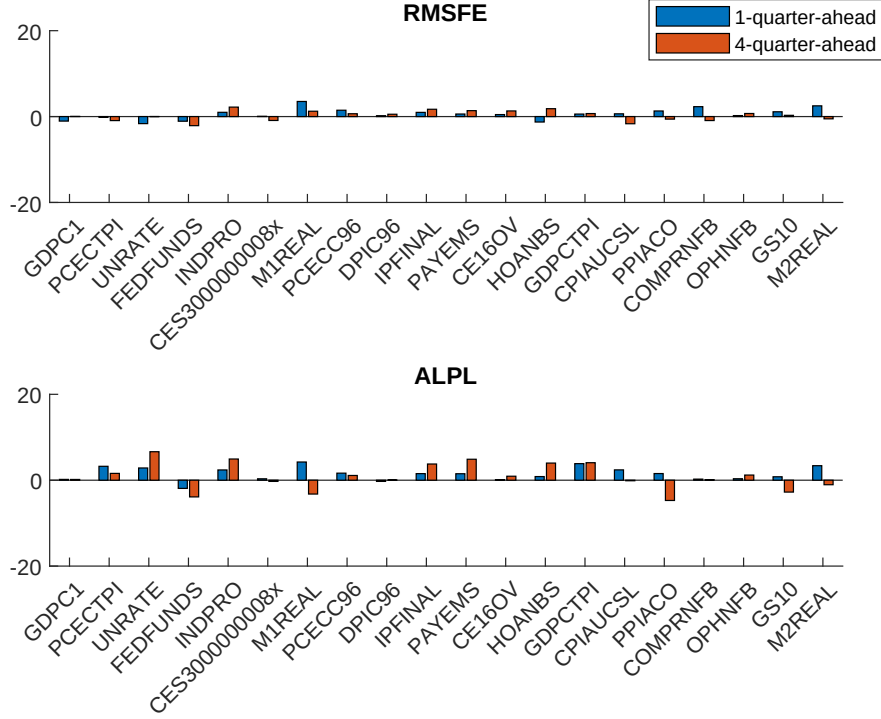


Figure 3: Forecasting results of an extension of HYB where the innovations to $h_{i,t}$ are correlated across equations compared to the benchmark HYB. The top panel shows the percentage gains in root mean squared forecast error of the extension. The symbols * and # after the mnemonic indicate rejection of equal forecast accuracy at significance level 0.05 using the test in Diebold and Mariano (1995) for 1- and 4-quarter-ahead point forecasts, respectively. The bottom panel presents the percentage gains in the average of log predictive likelihoods.

Next, we investigate the variability of point and density forecasts of 6 variables—real GDP, PCE inflation, unemployment, Fed funds rate and industrial production and real average hourly earnings in manufacturing—from the proposed model. We consider all 720 possible variable orderings and the setup of the forecasting exercise is described in Section 5.4 in the main text. For each variable ordering, we compute the root mean squared forecast error and the average of log predictive likelihoods from the proposed model as well as the HYB-(1, 1). We normalize the two metrics by those of the benchmark

ordering, and the results are reported in Figures 4-5.

Consistent with the results in Arias, Rubio-Ramirez, and Shin (2021), the variability of point forecasts is relatively small. Moreover, the variability of both point and density forecasts from the proposed model is similar to those in HYB-(1,1), a structural-form parameterization of the model in Primiceri (2005).

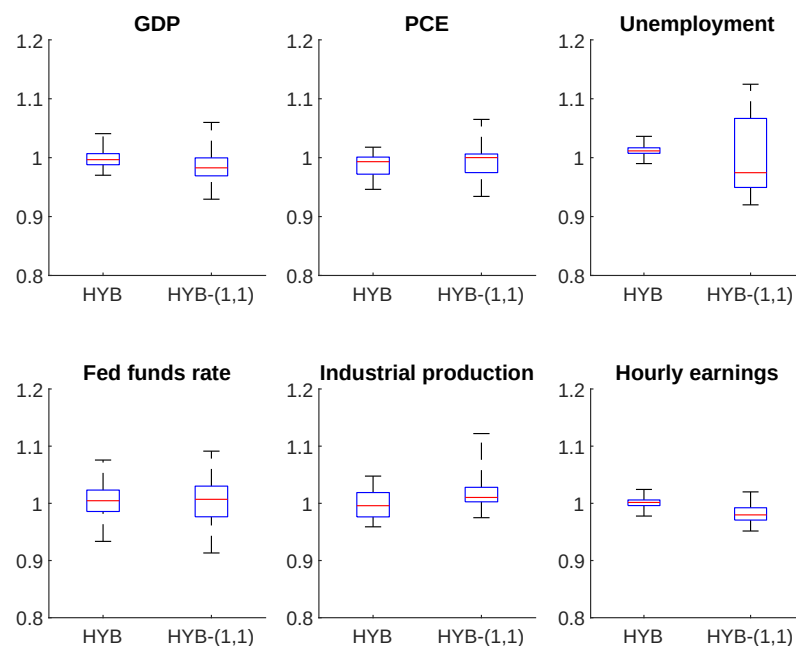


Figure 4: Boxplots of the relative mean squared forecast errors of 1-quarter-ahead point forecasts from the proposed hybrid TVP-VAR (HYB) and the full-fledged TVP-VAR (HYB-(1,1)).

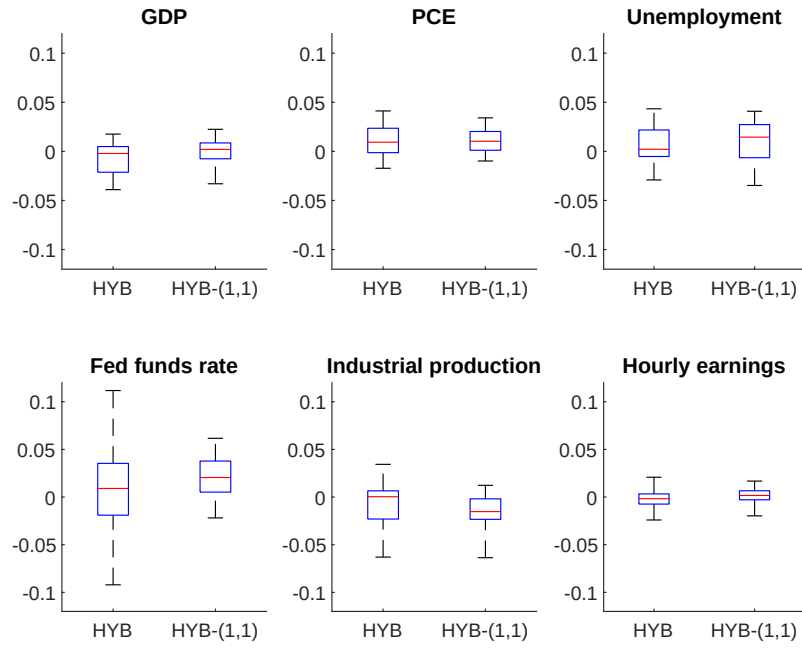


Figure 5: Boxplots of the relative average log predictive likelihoods of 1-quarter-ahead density forecasts from the proposed hybrid TVP-VAR (HYB) and the full-fledged TVP-VAR (HYB-(1,1)).

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